International Institute of Information Technology, Bangalore
CSEP 511 Data Structures and Algorithms Lab on 12th Jan 2026

1. Given $x, n, m, 2 \leq x \leq 10^9, 2 \leq m \leq 10^6$ find x^n mod m.
   (a) $n < 10^{18}$
   (b) $n < 2^{10^5}$, given in binary.
   (c) $n < 10^{10^5}$, given in decimal.
2. Given $A, n, m, 2 \leq m \leq 10^6, A$ is a $k * k, 0 < k < 100$ matrix, find A^n mod m.
   (a) $n < 10^{18}$
   (b) $n < 2^{10^3}$, given in binary.
   (c) $n < 10^{10^3}$, given in decimal.
3. (a) Given 10^5 bit binary number convert it to a decimal number.
   (b) Given 10^5 digit decimal number convert it to a binary number.
4. (a) Given two 10^5 bit binary numbers, add them.
   (b) Given two 10^5 digit decimal numbers, multiply them.
5. Find x mod m, where x is 10^5 digit number and m is an integer $1 < m < 10^6$